

COURSE 3 - DATA ENGINEERING 2026**C3-063 - CourierSpark Mini-Lab**

Integrated limited-hint Spark assessment on fresh shipment/route data.

Artifact	C3-063
Status	2026 - LEARNER EDITION
Phase	C3-05 Distributed Processing with Spark 16 lessons 46-60 focused hours
Prerequisite	C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here.
Current lane	Apache Spark / PySpark 4.2.0 Python 3.10+ Java 17+ learner local[2]
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

Scenario

CourierSpark receives shipment events and route metadata. Route R1 is intentionally much hotter than the others, and one shipment has missing `delivery_minutes`. Build a correct, tested Spark transformation and diagnose physical behavior without tuning by superstition.

Required build

- Define explicit schemas and read both inputs.
- Create clean + rejected/flagged outputs with a stated rule for missing `delivery_minutes`.
- Prove `route_id` uniqueness before joining.
- Compute route-level `shipment_count`, `delivered_count` and average `delivery_minutes`.
- Add a per-route delivery rank/window metric without changing shipment grain.
- Inspect partition counts and formatted physical plan; identify at least one Exchange/shuffle and the join operator.
- Measure route frequency; explain whether R1 is meaningful skew and what you would monitor at scale.
- Write curated Parquet with a sensible partition/file strategy.
- Add at least two PySpark tests/invariants.
- Package reusable functions/CLI and a short handoff/runbook.

Pass standard - competency evidence

PASS only when every required competency is demonstrated and zero unresolved critical failures remain.

Limited hints

- Use built-in Spark SQL functions; do not collect rows to Python for transformation.
- A tiny local dataset may not show dramatic timing differences. Plan/distribution evidence is sufficient.
- Do not salt a key merely because it is the largest; diagnose business validity and actual imbalance first.